E-commerce Order Fulfillment and Sales Performance Analysis

Power BI Dashboard Development Report

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Project: E-Commerce Retail Sales Analysis (group internship project)

Dataset: ecommerce_retail_transactions_cleaned.csv (12,000 order records)

Tool: Microsoft Power BI Desktop

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1. Introduction and Objective

This report describes how the Power BI dashboard for the project was built, starting from the import of the cleaned dataset and ending with four interactive report pages. It explains which business problem each page answers, which filters (slicers), KPI cards, DAX formulas and charts were used, why they were chosen, and what the dashboard shows.

The project was carried out by a team of six members. Member 2 prepared and cleaned the data with Python, Member 3 wrote the SQL queries and calculated the KPIs, Member 4 performed exploratory data analysis, and Member 5 (the author of this report) built the dashboard. The dashboard uses the cleaned dataset supplied by the team and was validated against the SQL KPI results.

The project brief proposes a delivery and logistics analysis. The brief also states that the analysis must use only the fields available in the dataset and must not fabricate missing data. The available dataset has no carrier, dispatch date, promised date or delivery date. For this reason, on-time delivery rate, delay and processing time cannot be calculated. The dashboard therefore focuses on order fulfillment (order status, cancellations and returns), sales performance and customer ratings, and it documents the missing fields as limitations. The dashboard is titled Order Fulfillment and Sales Performance Analysis so that its title matches what the data supports.

Objectives

- Build one consistent, trusted view of order volume, sales, customers, cancellations, returns and ratings.
- Compare performance by country, city, product category, year, quarter, payment method and weekday.
- Define every KPI clearly, state its denominator, and validate it against the source data.
- Present findings in plain language and separate observed patterns from proven causes.

2. Business Problems

The dashboard is organised around five business questions. Each report page answers one of them.

No.	Business problem	Answered on
1	How is the business performing overall? How many orders were placed, how much was sold, and to how many customers?	Page 1: Executive Overview
2	Where does the sales value come from: which countries, cities, product categories and years?	Page 2: Sales and Region
3	Where is order value lost through cancelled and returned orders, and are some segments worse than others?	Page 3: Cancellations and Returns
4	Are customers satisfied with their orders, and does satisfaction differ by segment or over time?	Page 4: Customer Experience

No.	Business problem	Answered on
5	What can the data not tell us?	Data notes on the pages and Section 9

3. Dataset Description

The dataset is a CSV file with 12,000 order records and 18 columns. Every Order_ID appears once, so each row is one order. Orders run from 1 January 2024 to 6 December 2026. The data covers 7 countries, 23 shipping cities, 8 product categories, 6 payment methods and 5 order statuses.

Column	Type	Meaning and use
Order_ID	Text	Unique order identifier. Used to count orders.
Customer_ID	Text	Customer identifier. Used to count purchasing customers.
Order_Date	Date	Date the order was placed. Linked to the date table.
Product_Category	Text	One of 8 categories (for example Electronics).
Product_Name	Text	Name of the product ordered.
Quantity	Whole number	Units ordered.
Unit_Price_USD	Decimal	Price per unit in US dollars.
Discount_Percent	Decimal	Discount given on the order line.
Payment_Method	Text	One of 6 methods (Credit Card, Debit Card, UPI, PayPal, Net Banking, Cash on Delivery).
Shipping_City	Text	Destination city. 494 orders have the value Unknown.
Country	Text	Destination country (7 countries).
Order_Status	Text	Delivered, Shipped, Pending, Cancelled or Returned.
Customer_Rating	Decimal	Rating from 1 to 5. Present only for delivered orders (5,944 orders).
Flag_Future_Date, Flag_Zero_Quantity, Flag_Negative_Quantity, Flag_Negative_Status_Mismatch	True/False	Four data-quality flags added during cleaning. 217 orders (1.8 percent) carry at least one flag.
Provisional_Line_Value_USD	Decimal	Provisional value of the order line in US dollars. Used as the sales value.

Order status distribution

Status	Orders	Share of 12,000
Delivered	6,578	54.8%
Shipped	1,816	15.1%
Pending	1,395	11.6%
Cancelled	1,196	10.0%
Returned	1,015	8.5%

Fields not available in the dataset

Carrier, dispatch date, promised delivery date, actual delivery date, distance, product cost, shipping cost, operating cost and cancellation reason are not in the dataset. These fields were not created or estimated.

4. Data Import and Preparation in Power BI

4.1 Importing the data

- Opened Power BI Desktop and used Home, Get data, Text/CSV to select the cleaned CSV file.
- Checked the preview: first row used as column names, file origin UTF-8, delimiter comma.
- Chose Transform Data instead of Load, so the data could be inspected and typed in Power Query before use.
- Renamed the table to Orders. All DAX formulas refer to Orders, so a short, fixed name keeps them readable and consistent.

4.2 Setting data types

Correct types are needed so that dates form a time axis and numbers can be summed. The types used were: Text for identifiers, category, city, country, payment method and status; Date for Order_Date; Whole Number for Quantity; Decimal Number for prices, discount, rating and line value; True/False for the four flag columns.

4.3 Handling missing ratings

Customer_Rating is blank for 6,056 orders because only delivered orders can be rated. These blanks were kept as blanks. A blank means the customer did not rate the order, so it must not be treated as a rating of zero. Replacing the blanks with zero would pull the average rating from 3.90 down to about 1.93 and would count unrated orders as rated.

4.4 Keeping all rows and marking invalid ones

No rows were deleted, in line with the brief, which asks that unusual records are not removed without a documented reason. Instead, a calculated column named Is_Valid was added. It equals 0 when any of the four flags is true and 1 otherwise. KPI measures use Is_Valid inside the formula, so the excluded records are visible and the exclusion is reversible.

```
Is_Valid =
IF(
    Orders[Flag_Future_Date] || Orders[Flag_Zero_Quantity]
    || Orders[Flag_Negative_Quantity] || Orders[Flag_Negative_Status_Mismatch],
    0, 1
)
```

Check: the sum of Is_Valid is 11,783, which equals 12,000 minus 217 flagged records.

4.5 Date table and relationship

A separate date table was created so that years, quarters, months and weekdays can be used on charts and slicers.

```
DateTable =
ADDCOLUMNS(
    CALENDAR(DATE(2024,1,1), DATE(2026,12,31)),
    "Year", YEAR([Date]),
    "Month No", MONTH([Date]),
    "Month", FORMAT([Date], "MMM"),
    "Year-Month", FORMAT([Date], "YYYY-MM"),
    "Weekday No", WEEKDAY([Date], 2),
    "Weekday", FORMAT([Date], "ddd")
)
```

- Month was sorted by Month No and Weekday by Weekday No, so charts follow calendar order (Monday to Sunday) and not alphabetical order.
- Summarization was set to Do not summarize for Year, Month No and Weekday No, so these numbers are never added together.
- The table was marked as a date table.
- A one-to-many relationship was created from DateTable[Date] to Orders[Order_Date] with single cross-filter direction. Without this relationship, every quarter showed the same total value, because the date table could not filter the orders.

4.6 Helper columns

```
Year-Quarter = YEAR([Date]) & " Q" & QUARTER([Date])

City_Known = IF(Orders[Shipping_City] = "Unknown", "Unknown", "Known")

Star Rating =
IF(Orders[Customer_Rating] >= 1,
    FORMAT(Orders[Customer_Rating], "0") & " star")
```

- Year-Quarter (in DateTable) is the axis of the quarterly trend charts.
- City_Known is used as a visual filter so that the Unknown city is excluded from the Top 10 Cities ranking.

- Star Rating turns the numeric rating into labels such as 5 star for the rating distribution chart.

4.7 Page size and colours

Each page uses a fixed canvas of 1280 by 720 pixels. All elements were placed with exact position values so the four pages share one grid. The colour palette is described in Section 6.

5. KPI Definitions and DAX Formulas

A measure is a formula that recalculates whenever a slicer changes. All measures below work only on the imported Orders table. Where a status is named inside the formula, that status is fixed for the measure, and slicers on Order Status do not change it.

5.1 Sales and order measures

```
Total Orders = DISTINCTCOUNT(Orders[Order_ID])
```

Counts every placed order once, whatever its status. It is the denominator of the cancellation and return rates. Value: 12,000.

```
Eligible Delivered Orders =
CALCULATE(
    DISTINCTCOUNT(Orders[Order_ID]),
    Orders[Order_Status] = "Delivered",
    Orders[Is_Valid] = 1
)
```

Counts delivered orders that passed the data-quality checks. Delivered orders with a flag are excluded. Value: 6,451 (6,578 delivered orders minus 127 that carry a flag).

```
Total Sales =
CALCULATE(
    SUM(Orders[Provisional_Line_Value_USD]),
    Orders[Order_Status] = "Delivered",
    Orders[Is_Valid] = 1
)
```

Adds the line value of eligible delivered orders. Only delivered orders count as realised sales. Cancelled, returned, shipped and pending orders are not counted. Value: about $764,592.

```
Purchasing Customers =
CALCULATE(
    DISTINCTCOUNT(Orders[Customer_ID]),
    Orders[Order_Status] = "Delivered",
    Orders[Is_Valid] = 1
)
```

Counts customers who have at least one eligible delivered order. Value: 2,935.

```
Avg Order Value = DIVIDE([Total Sales], [Eligible Delivered Orders])
```

Average sales value per eligible delivered order. DIVIDE returns a blank instead of an error when the denominator is zero, for example after a filter with no orders. Value: $118.52.

5.2 Cancellation and return measures

```
Cancellation Rate =
DIVIDE(
    CALCULATE(DISTINCTCOUNT(Orders[Order_ID]), Orders[Order_Status] = "Cancelled"),
    DISTINCTCOUNT(Orders[Order_ID])
)
```

Cancelled orders divided by all placed orders. The denominator is all orders, as the brief asks the denominator of each rate to be stated. Value: 9.97 percent (1,196 of 12,000).

```
Return Rate =
DIVIDE(
    CALCULATE(DISTINCTCOUNT(Orders[Order_ID]), Orders[Order_Status] = "Returned"),
    DISTINCTCOUNT(Orders[Order_ID])
)
```

Returned orders divided by all placed orders. Value: 8.46 percent (1,015 of 12,000).

```
Cancelled Orders =
CALCULATE(DISTINCTCOUNT(Orders[Order_ID]), Orders[Order_Status] = "Cancelled")

Returned Orders =
CALCULATE(DISTINCTCOUNT(Orders[Order_ID]), Orders[Order_Status] = "Returned")

Revenue at Risk =
CALCULATE(
    SUM(Orders[Provisional_Line_Value_USD]),
    Orders[Order_Status] IN {"Cancelled", "Returned"}
)
```

Revenue at Risk is the order value of cancelled and returned orders: $138,782.58 plus $112,362.34, a total of $251,144.92. It shows the size of the value lost or exposed. It is based on the provisional line value, not on profit.

5.3 Rating measures

```
Avg Rating = AVERAGE(Orders[Customer_Rating])
```

Average of the ratings given. AVERAGE ignores blank cells, so only the 5,944 rated orders are used. Value: 3.90.

```
Rated Orders =
CALCULATE(DISTINCTCOUNT(Orders[Order_ID]),
NOT(ISBLANK(Orders[Customer_Rating])))
```

Number of orders that have a rating. It is also the denominator of the star-share measures. Value: 5,944.

```
Star Share 4-5 =
DIVIDE(
    CALCULATE(DISTINCTCOUNT(Orders[Order_ID]), Orders[Customer_Rating] >= 4),
    [Rated Orders]
)

Star Share 1-2 =
DIVIDE(
    CALCULATE(DISTINCTCOUNT(Orders[Order_ID]),
        Orders[Customer_Rating] >= 1, Orders[Customer_Rating] <= 2),
    [Rated Orders]
)
```

Share of rated orders with 4 or 5 stars (70.4 percent) and with 1 or 2 stars (12.4 percent). The condition Customer_Rating >= 1 in the second measure is needed because a blank rating would otherwise be treated as less than or equal to 2 and the share would be overstated.

5.4 Display measures for cards

Power BI cards shorten large numbers (for example 6,451 becomes 6.451K). To show the full number, text versions of the measures were created with FORMAT and used only in card visuals. Charts use the numeric measures, because text cannot be plotted or calculated.

```
Total Orders Display = FORMAT([Total Orders], "#,##0")
Total Sales Display = FORMAT([Total Sales], "$#,##0")
Rated Orders Display = FORMAT([Rated Orders], "#,##0")
```

5.5 Summary of KPI results

KPI	Value	Denominator or basis
Total Orders	12,000	All placed orders
Eligible Delivered Orders	6,451	Delivered, no data-quality flag
Total Sales	$764,592	Eligible delivered orders
Purchasing Customers	2,935	Customers with an eligible delivered order
Average Order Value	$118.52	Total Sales divided by eligible delivered orders
Cancellation Rate	9.97%	Cancelled orders divided by 12,000

KPI	Value	Denominator or basis
Return Rate	8.46%	Returned orders divided by 12,000
Revenue at Risk	$251,144.92	Value of cancelled and returned orders
Average Rating	3.90	Orders with a rating (5,944)
4 to 5 Star Share	70.4%	Rated orders
1 to 2 Star Share	12.4%	Rated orders

6. Dashboard Design and Standards

All four pages follow the same layout and colours so that the dashboard looks like one product. A dark navy header carries the project title and the page name. Four slicers sit below it, KPI cards come next, charts follow in a two-by-two grid, and a Key Insights box closes each page.

Colour	Hex code	Use
Deep Navy	#102F4A	Header and main titles
Dark Blue	#173E5C	Slicers and table headers
Teal	#07858C	Main chart colour (sales, trends, ratings)
Turquoise	#12A0A5	Secondary chart elements
Royal Blue	#2878D0	Comparisons and rankings
Sky Blue	#67A9E0	Supporting chart elements
Golden Orange	#F5A623	Key Insights box border and highlights only
Light Orange	#FFBD59	Secondary highlights
Red	#D94B45	Cancellations and negative metrics only
Green	#329B78	Positive indicators
White / Light Blue Grey	#FFFFFF / #EDF3F7	Cards / page background
Dark Text	#263746	Titles, labels and descriptions

Design rules

- One colour carries one meaning on every page: teal for main metrics, blue for rankings, orange for insights, red for negative results.
- Bar and column axes start at zero. Axes were not shortened to exaggerate small differences.
- Charts use one colour each. The default alternating colours were switched off.
- Chart titles state the measure and the scope. Axis titles that repeat the chart title were removed.
- The year, country and product category slicers are synchronised across all pages, so a filter chosen on one page also applies to the others.

7. Page-by-Page Description

7.1 Page 1: Executive Overview

Business question: How is the business performing overall?

Slicers used

Slicer	Field	Why
Year	DateTable[Year]	Filters the page to one or more years.
Country	Orders[Country]	Filters to one or more countries.
Product Category	Orders[Product_Category]	Filters to one or more categories.
Order Status	Orders[Order_Status]	Filters order counts by status. It affects only Total Orders and the Orders by Status chart, because the sales measures already fix the status to Delivered.

KPI cards

Card	Measure	Value	Why this KPI
Total Orders	Total Orders Display	12,000	Size of the order base.
Eligible Delivered	Eligible Delivered Display	6,451	Orders that became realised, valid sales.
Total Sales	Total Sales Display	$764,592	Revenue from delivered orders.
Purchasing Customers	Purchasing Customers Display	2,935	Number of buying customers.
Avg Order Value	Avg Order Value	$118.52	Spend per order.
Cancellation Rate	Cancellation Rate	9.97%	Main fulfillment-failure indicator. Shown in red.

Visuals

Visual	Type and fields	Why this chart
Quarterly Sales Trend	Column chart. X: DateTable[Year-Quarter]. Y: Total Sales. 2026 Q3 and Q4 excluded.	Shows how sales move over time. The two excluded quarters hold only 52 and 47 orders in total, so they are incomplete.
Orders by Status	Donut chart. Legend: Order_Status. Values: Total Orders.	The five statuses are parts of one total of 12,000 orders, so a donut shows their shares.

Output and findings

- Delivered sales are $764.6K from 6,451 valid orders placed by 2,935 customers, about 2.2 orders per customer. The average order value is $118.52.

- Of all orders, 54.8 percent are delivered, 26.8 percent are still shipped or pending, 10.0 percent are cancelled and 8.5 percent are returned.
- The 2,211 cancelled or returned orders (18.4 percent) carry about $251K of order value, roughly one third of delivered sales.
- Quarterly sales were between $74.8K and $81.6K from 2024 Q1 to 2025 Q4 ($75.4K, $81.6K, $78.2K, $74.9K, $74.8K, $76.8K, $77.2K, $80.0K). They were $69.1K in 2026 Q1 and $71.6K in 2026 Q2, about 7 percent below the same quarters of 2025.
- Context: orders placed fell about 3.5 percent year on year in these quarters (1,108 against 1,150 in Q1, and 1,164 against 1,206 in Q2), while the delivered share stayed near 55 percent. The data cannot explain the rest of the drop.

The donut shows 6,578 delivered orders while the KPI card shows 6,451 eligible delivered orders. The donut counts every delivered order. The card removes the 127 delivered orders that carry a data-quality flag. This difference is stated in the data note on the page.

Key Insights text on the page

- Delivered sales: $764.6K from 6,451 valid orders; average order value $118.52.
- 18.4 percent of orders (2,211) were cancelled or returned, about $251K of order value.
- Quarterly sales held near $75K to $82K in 2024 and 2025 and eased to about $70K in 2026 H1.
- 2026 Q3 and Q4 are excluded from the trend: only about 50 orders each.

7.2 Page 2: Sales and Region

Business question: Where does the sales value come from?

Slicers used: Year, Country, Product Category and Order Status, the same as Page 1, so the filters stay familiar and synchronised. The same six KPI cards sit at the top of the page, so that the viewer always sees the totals that the charts split up.

Visuals

Visual	Type and fields	Why this chart
Sales by Country	Column chart. X: Country. Y: Total Sales.	Compares seven countries side by side.
Top 10 Cities by Sales	Horizontal bar chart. Y: Shipping_City. X: Total Sales. Filters: Top N 10 by Total Sales, and City_Known = Known.	A ranking with long names reads best as horizontal bars. The Unknown city is not a real place, so it is excluded.
Sales by Product Category	Treemap. Group: Product_Category. Values: Total Sales.	Shows each category as a share of total sales. Size and colour show the large share of Electronics.
Sales by Year	Donut chart. Legend: Year. Values: Total Sales.	Three years are parts of one total. The page notes that 2026 is a partial year.

Output and findings

Country	Sales	Top 10 city	Sales
United Arab Emirates	$117.2K	Abu Dhabi	$61.3K
United Kingdom	$114.2K	Dubai	$51.6K
Germany	$113.0K	Hamburg	$41.1K
India	$110.8K	Birmingham	$37.7K
United States	$106.8K	London	$37.2K
Canada	$106.1K	Toronto	$35.6K
Australia	$96.6K	Melbourne	$35.5K
		Munich	$35.4K
		Montreal	$33.6K
		Manchester	$32.7K

Product category	Sales	Share	Year	Sales	Share
Electronics	$203.0K	26.6%	2024	$310.1K	40.6%
Home and Kitchen	$151.2K	19.8%	2025	$308.9K	40.4%
Sports	$123.7K	16.2%	2026 (partial)	$145.6K	19.0%
Fashion	$99.7K	13.0%			
Toys	$77.5K	10.1%			
Beauty	$46.9K	6.1%			
Books	$36.7K	4.8%			
Grocery	$25.9K	3.4%			

- Electronics is the largest category with about 27 percent of sales. The top three categories (Electronics, Home and Kitchen, Sports) together make up 62.5 percent.
- Sales are spread fairly evenly over the seven countries. The lowest country (Australia) reaches about 82 percent of the highest (the UAE).
- Abu Dhabi and Dubai lead the city ranking and together contribute about $113K.
- Sales were about $310K in 2024 and $309K in 2025. 2026 holds $146K because the year is incomplete (only about 50 orders in each of Q3 and Q4), so it must not be read as a decline.
- 270 delivered orders have an Unknown city. They are left out of the city ranking and noted on the page.

Key Insights text on the page

- Electronics is the largest category at $203K (about 27 percent of sales); the top three categories make up about 62 percent.

- Sales are spread evenly across countries ($97K Australia to $117K UAE); Abu Dhabi ($61K) and Dubai ($52K) lead the cities.
- Sales were about $310K in 2024 and $309K in 2025; 2026 is a partial year, so it is not comparable.
- Top 10 Cities excludes 270 delivered orders with an unknown city.

7.3 Page 3: Cancellations and Returns

Business question: Where is order value lost, and are some segments worse than others?

Slicers used

Slicer	Field	Why
Year	DateTable[Year]	Filters to one or more years.
Country	Orders[Country]	Filters to one or more countries.
Product Category	Orders[Product_Category]	Filters to one or more categories.
Payment Method	Orders[Payment_Method]	Replaces the Order Status slicer on this page. The cancelled and returned measures are built on status, so a status slicer would reduce them to zero and confuse the viewer.

KPI cards

Card	Measure	Value
Cancellation Rate	Cancellation Rate	9.97%
Return Rate	Return Rate	8.46%
Revenue at Risk	Revenue at Risk Display	$251,144.92
Cancelled Orders	Cancelled Orders Display	1,196
Returned Orders	Returned Orders Display	1,015

Visuals

Visual	Type and fields	Why this chart
Cancellation % by Category	Column chart. X: Product_Category. Y: Cancellation Rate.	Compares the cancellation rate across 8 categories. Red marks a negative metric.
Return % by Country	Horizontal bar chart. Y: Country. X: Return Rate.	Compares the return rate across 7 countries.
Cancellation % by Payment Method	Column chart. X: Payment_Method. Y: Cancellation Rate.	Checks whether a payment method is linked to more cancellations.

Visual	Type and fields	Why this chart
Cancellation % by Weekday	Column chart. X: DateTable[Weekday] (Monday to Sunday). Y: Cancellation Rate.	Checks for a weekday pattern. The weekday sort keeps the days in calendar order.

Output and findings

Cancellation by category	Rate	Cancellation by payment	Rate
Fashion	10.50%	PayPal	10.87%
Grocery	10.33%	Net Banking	10.32%
Home and Kitchen	10.29%	Cash on Delivery	10.05%
Sports	10.08%	Credit Card	9.85%
Electronics	10.05%	Debit Card	9.48%
Beauty	9.91%	UPI	9.17%
Toys	9.31%		
Books	9.31%		

Cancellation by weekday	Rate	Return by country	Rate
Monday	10.71%	Canada	9.24%
Tuesday	10.36%	United Kingdom	8.60%
Wednesday	9.18%	Australia	8.55%
Thursday	9.61%	United States	8.47%
Friday	9.44%	India	8.22%
Saturday	9.73%	United Arab Emirates	8.15%
Sunday	10.73%	Germany	8.01%

- The cancelled and returned orders carry $251,144.92: $138,782.58 from cancelled orders and $112,362.34 from returned orders. This is 18.4 percent of all orders and about one third of delivered sales.
- Cancellation rates are between 9.2 percent and 10.9 percent in every category, payment method and weekday. Return rates by country are between 8.0 percent and 9.2 percent.
- The highest values are PayPal (10.87 percent), Fashion (10.50 percent) and Sunday and Monday (about 10.7 percent). The gaps to the lowest values are small and may be random variation. No segment stands out as a clear hotspot.
- The dataset has no cancellation or return reasons, so causes cannot be determined.

Key Insights text on the page

- Cancellation is about 9 to 11 percent in every category, payment method and weekday; the return rate by country is about 8 to 9 percent.

- The highest cancellation rates are Fashion (10.5 percent), PayPal (10.9 percent) and Sunday and Monday (10.7 percent), but the gaps are small and may be random.
- Cancelled and returned orders (1,196 and 1,015) carry about $251K of order value.
- Cancellation reasons are not in the dataset, so causes cannot be determined.

7.4 Page 4: Customer Experience

Business question: Are customers satisfied with their orders?

Slicers used: Year, Country, Product Category and Payment Method, the same as Page 3. An Order Status slicer is not used, because ratings exist only for delivered orders.

KPI cards

Card	Measure	Value	Why this KPI
Avg Rating	Avg Rating	3.90	Overall satisfaction on a 1 to 5 scale.
Rated Orders	Rated Orders Display	5,944	Sample size behind every rating figure.
4 to 5 Star Share	Star Share 4-5	70.4%	Share of satisfied customers.
1 to 2 Star Share	Star Share 1-2	12.4%	Share of dissatisfied customers. Shown in red.

Visuals

Visual	Type and fields	Why this chart
Rating Distribution	Donut chart. Legend: Star Rating. Values: Rated Orders. Blank ratings filtered out.	The five star levels are parts of one total of 5,944 rated orders.
Avg Rating by Category	Horizontal bar chart. Y: Product_Category. X: Avg Rating. Axis from 0 to 5.	Compares the average rating across categories. The axis starts at zero so small differences are not exaggerated.
Rating by Country	Table with data bars. Columns: Country, Rated Orders, Avg Rating.	Shows the number of ratings behind each average, so sample size is visible.
Avg Rating Trend by Quarter	Line chart. X: Year-Quarter. Y: Avg Rating. Axis from 0 to 5. 2026 Q3 and Q4 excluded.	Shows whether satisfaction changes over time. The two excluded quarters hold only 28 and 24 ratings.

Output and findings

Stars	Orders	Share	Country	Rated orders	Avg rating
5 star	2,153	36.2%	Canada	817	3.95
4 star	2,031	34.2%	Australia	788	3.94

Stars	Orders	Share	Country	Rated orders	Avg rating
3 star	1,025	17.2%	India	852	3.91
2 star	475	8.0%	United Kingdom	895	3.90
1 star	260	4.4%	United Arab Emirates	855	3.90
Total	5,944	100%	United States	844	3.90
			Germany	893	3.81

- The average rating is 3.90 across 5,944 rated orders. 70.4 percent of ratings are 4 or 5 stars and 12.4 percent are 1 or 2 stars.
- Average rating by category runs from 3.85 (Books) to 4.00 (Electronics): Electronics 4.00, Home and Kitchen 3.91, Sports 3.90, Fashion 3.90, Beauty 3.89, Grocery 3.87, Toys 3.87, Books 3.85.
- Average rating by country runs from 3.81 (Germany) to 3.95 (Canada). Each country has between 788 and 895 ratings, so the samples are similar in size.
- Quarterly average rating stays between 3.82 and 3.96 from 2024 Q1 to 2026 Q2 (3.89, 3.89, 3.95, 3.89, 3.96, 3.82, 3.90, 3.92, 3.87, 3.90), so satisfaction is stable.
- Ratings exist only for delivered orders and the data has no delivery dates, so the comparison of ratings for on-time and late orders asked for in the brief cannot be made.

Key Insights text on the page

- Average rating is 3.90 across 5,944 rated orders; 70.4 percent are 4 to 5 stars and 12.4 percent are 1 to 2 stars.
- Ratings are similar across categories (3.85 to 4.00) and countries (3.81 to 3.95); quarterly averages stay between 3.82 and 3.96.
- Ratings exist only for delivered orders and there are no delivery dates, so on-time versus late ratings cannot be compared.

8. Validation of Results

Every figure on the dashboard was checked before it was reported. Three checks were used: a recalculation from the CSV file in Python, a comparison with the SQL KPIs of Member 3 and the EDA report of Member 4, and a filter test.

Check	Result
Total Orders, Eligible Delivered Orders, Purchasing Customers, Average Order Value, Cancellation Rate, Return Rate, Average Rating	Match the CSV recalculation and the figures reported by Members 3 and 4.
Total Sales	Dashboard $764,591.99, team $764,592.24. The gap of $0.25 comes from rounding: the dashboard sums the stored line values, while the team recomputed each line as quantity times unit price times one minus the discount. Both show $764,592 in whole dollars.

Check	Result
Category and country sales, top cities, yearly sales	Match the CSV recalculation. Electronics $203,046 and the UAE $117,211 agree with the EDA report.
Cancellation and return rates by segment	All category, payment, weekday and country values match the CSV.
Rating figures	Distribution, category, country and quarterly averages match the CSV.
Filter test with Country set to India	Total Orders 1,690, eligible delivered 914, sales $110,752, 811 customers, average order value $121.17, cancellation rate 9.35 percent, and the status counts 938, 256, 199, 158 and 139 all match the CSV for India.

9. Limitations

- No carrier, dispatch, promised delivery or actual delivery dates. On-time delivery rate, average delay, delivery duration, processing time and carrier comparison cannot be calculated.
- No product cost, shipping cost or operating cost. Profit cannot be calculated. Sales are based on the provisional line value.
- No cancellation or return reasons. The dashboard shows where cancellations and returns occur, but not why.
- Ratings exist only for delivered orders, and they cannot be linked to delivery performance.
- 2026 is a partial year. 2026 Q3 and Q4 contain only about 50 orders each, so they are left out of the trend charts and 2026 is not compared with full years.
- 494 orders (4.1 percent) have an Unknown city. 270 of the delivered ones are left out of the city ranking.
- 217 orders (1.8 percent) carry a data-quality flag. They are excluded from the sales KPIs but kept in order counts and status charts. For this reason the donut shows 6,578 delivered orders and the KPI card shows 6,451.
- Differences between segments are small (for example cancellation rates of 9 to 11 percent). They describe observed patterns and are not proof of cause. No significance test has been applied.
- Key Insights text boxes are written for the unfiltered view (all years and countries) and do not change with the slicers.

10. Recommendations and Measures to Track

The following actions are hypotheses based on the observed patterns. They are not proven causes.

No.	Recommended action	Reason
1	Record carrier, dispatch date, promised date and delivery date for every order.	These fields are needed for the core logistics KPIs in the brief: on-time rate, delay and processing time.

No.	Recommended action	Reason
2	Record the reason for each cancellation and return.	The dashboard shows $251K of order value at risk, but without reasons no targeted action is possible.
3	Add product, shipping and operating cost.	This allows profit and margin analysis.
4	Improve capture of the shipping city.	4.1 percent of orders have an Unknown city.
5	Monitor cancellation and return rates monthly and investigate any segment that moves clearly above the current range of about 9 to 11 percent.	Today no segment stands out, so a sustained rise would be a meaningful signal.
6	When the data for the second half of 2026 is complete, review the 2026 sales level again.	In 2026 Q1 and Q2, sales were about 7 percent and orders about 3.5 percent below the same quarters of 2025.
7	Optionally test whether the small differences in cancellation rate between segments are statistically significant.	A chi-square test of independence would show whether the gaps could be random.

Measures to track after changes

Measure	Baseline	Source
Cancellation Rate	9.97%	Page 1 and Page 3
Return Rate	8.46%	Page 3
Revenue at Risk	$251,144.92	Page 3
Average Rating	3.90	Page 4
1 to 2 Star Share	12.4%	Page 4
Total Sales per quarter	About $70K to $80K	Page 1
On-time delivery rate and delay	Not measurable yet	Requires delivery date fields

11. Conclusion

The dashboard turns the cleaned order data into four connected pages that answer four business questions: how the business performs, where sales come from, where order value is lost, and how satisfied customers are. Every KPI has a stated definition and denominator, and every figure was checked against the source data and the team SQL results.

The main findings are delivered sales of $764.6K, an average order value of $118.52, an 18.4 percent share of orders lost to cancellation and return (about $251K of order value), evenly spread sales across countries with Electronics as the largest category, and stable customer ratings with an average of 3.90. Cancellation and return rates are similar in every segment, so the data does not point to one specific problem area.

The dataset has no carrier or delivery-date fields, so the logistics measures named in the brief could not be produced. The report states this openly and recommends collecting these fields so that on-time delivery and delay analysis can be added in a later version.